



Constraining the Photometric Redshifts of Quasars using Variability

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LSST AGN SC



SCAN TO GET THE PAPER!

Context

AGN photometric redshift estimation using only SED colors is highly degenerate due to smooth continuum, host galaxy contamination, variability, dust effects, and degeneracies in color–redshift space.

AGN variability is shaped by accretion physics and modulated by cosmological time dilation, making it inherently redshift dependent. VAR-PZ uses this property to derive a variability-based redshift likelihood, which, when combined with SED priors, significantly improves photo-z accuracy.

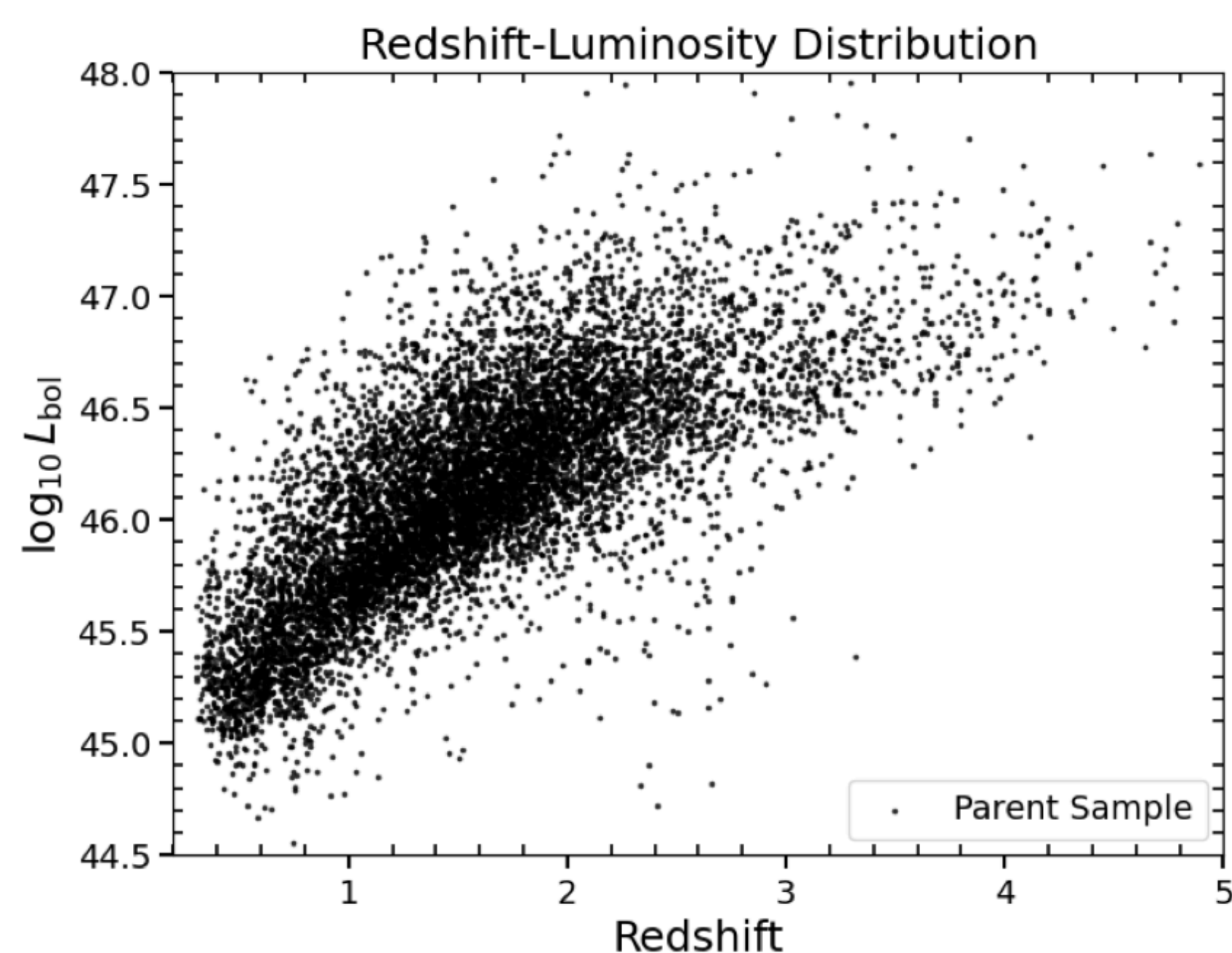


Fig. 1. Distribution of the sample quasars in the luminosity (bolometric) and redshift space

Sample selection

- 9210 spectroscopically confirmed quasars with SDSS light curves between redshift range of 0.3 to 5.0
- Simulated SDSS and LSST-like light curves

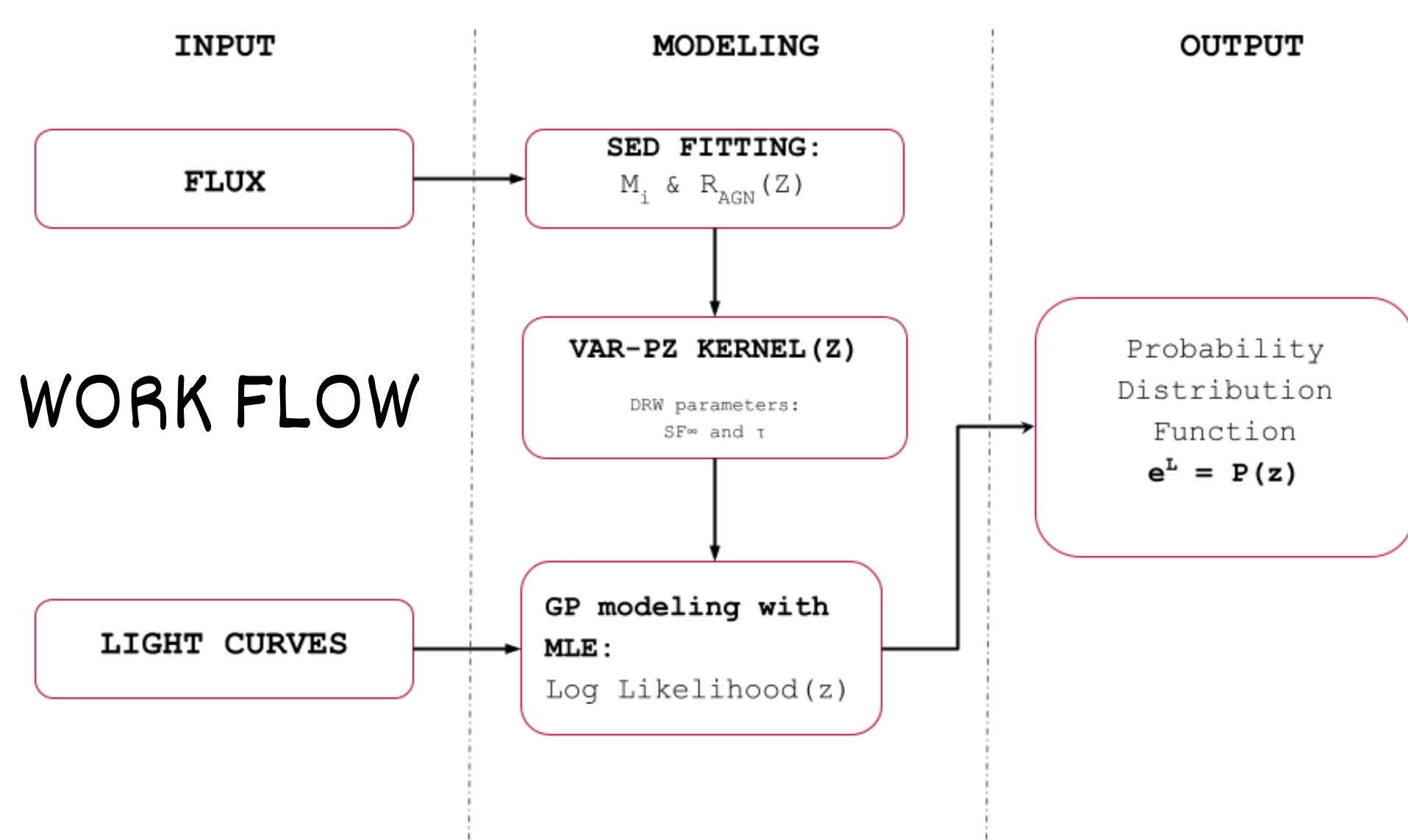
Goals and Methods

MODELING THE AGN VARIABILITY AS A DAMPED RANDOM WALK

$$\log_{10} f = A + B \log_{10} \left(\frac{\lambda_{\text{RF}}}{4000 \text{ \AA}} \right) + C (M_i + 23)$$

Methodology

- VAR-PZ models the AGN variability as DRW process as a function of redshift
- Uses CELERITE to model the light curves through gaussian process.
- M_i , AGN_flux fraction is obtained through SED modeling using Assef et al. 2010 templates.



f	A	$B (\lambda_{\text{RF}})$	$C (M_i)$	$D (M_{\text{BH}})$	Reference
SF_{∞}	-0.51 ± 0.02	-0.479 ± 0.005	0.131 ± 0.008	0.18 ± 0.03	M10
SF_{∞}	-0.618 ± 0.007	-0.479 ± 0.005	0.090 ± 0.003	0	
τ	2.4 ± 0.2	0.17 ± 0.02	0.03 ± 0.04	0.21 ± 0.07	
τ	2.2 ± 0.1	0.17 ± 0.02	-0.01 ± 0.02	0	
SF_{∞}	-0.695 ± 0.001	-0.456 ± 0.03	0.086 ± 0.001	0.012 ± 0.001	This Work
SF_{∞}	-0.713 ± 0.003	-0.456 ± 0.03	0.071 ± 0.002	0	
τ	2.2 ± 0.004	0.19 ± 0.01	-0.027 ± 0.002	0.027 ± 0.003	
τ	2.2 ± 0.019	0.19 ± 0.01	-0.026 ± 0.011	0	

Table 1.: Best-fit coefficients for the DRW from MacLeod+2010 and this work.

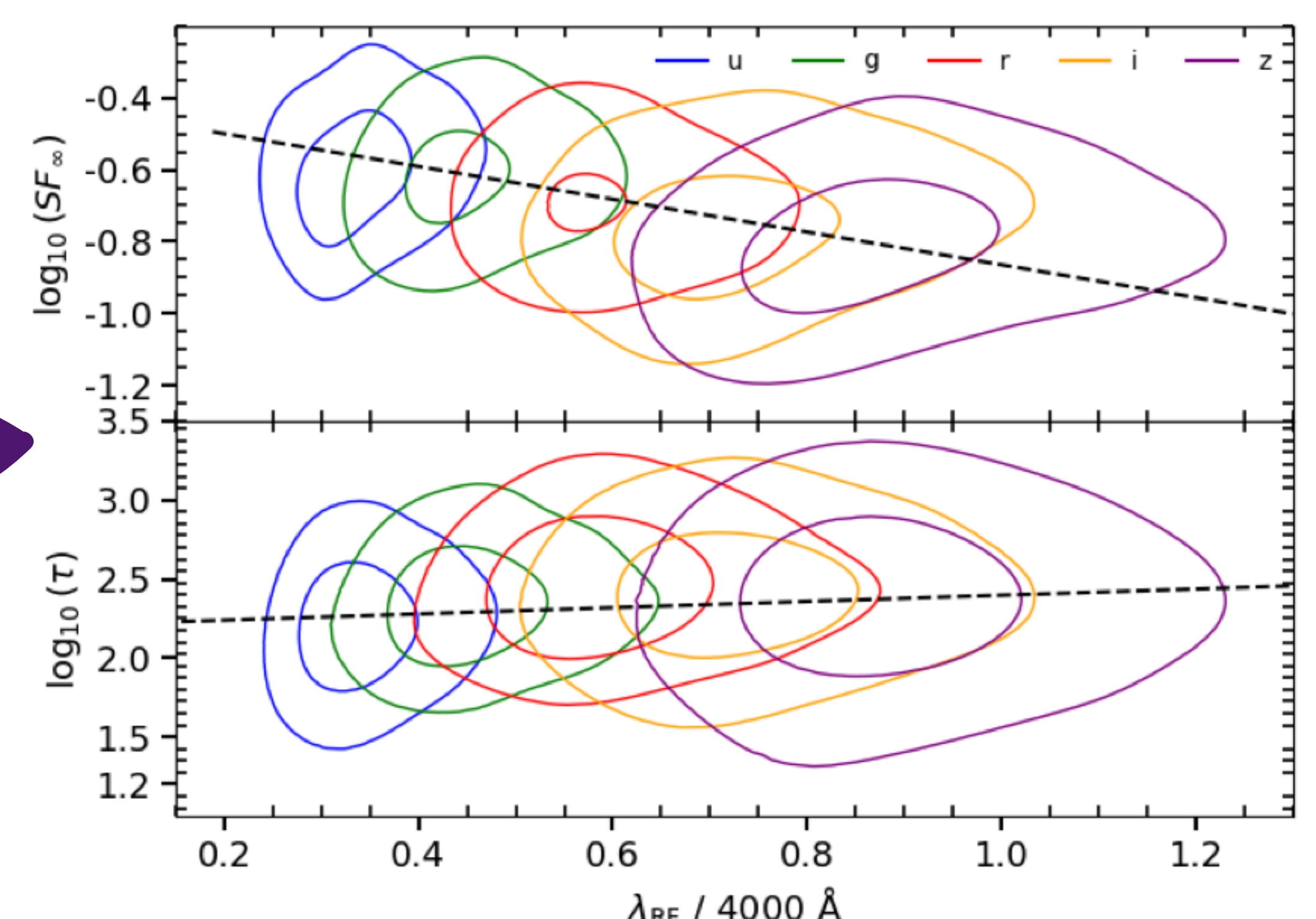


Fig. 2. The distribution of the variability parameters, SF_{∞} and τ , in the observed frame, measured from the ugriz bands as a function of rest frame wavelength.

Results

- Short baselines of SDSS limits the constraining power as ($\text{Baseline} > 3\tau$; Burke et al. 2021): 10 % reduction in the outlier fraction
- LSST like simulations show that VAR-PZ will be able to reduce the outlier fraction from 32% (SED-alone) to 15% (SED+Variability) at the end of the survey.

Application

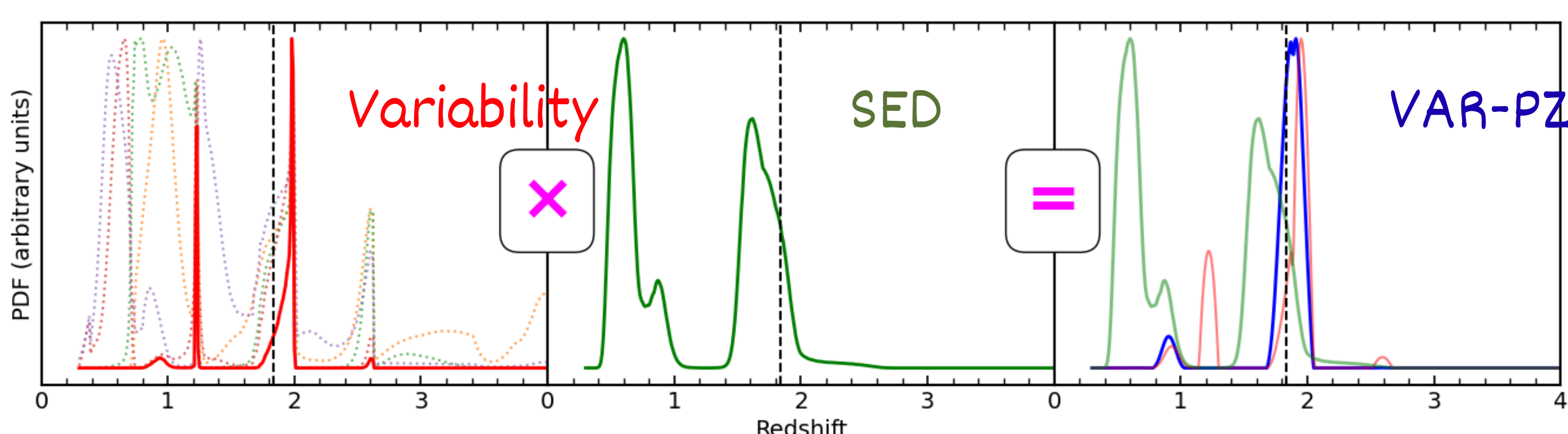


Fig. 4. Photo-z priors from VAR-PZ along with SED fitting

PDF, and the combined posterior. Vertical dashed lines mark the trial redshifts from the top panel.

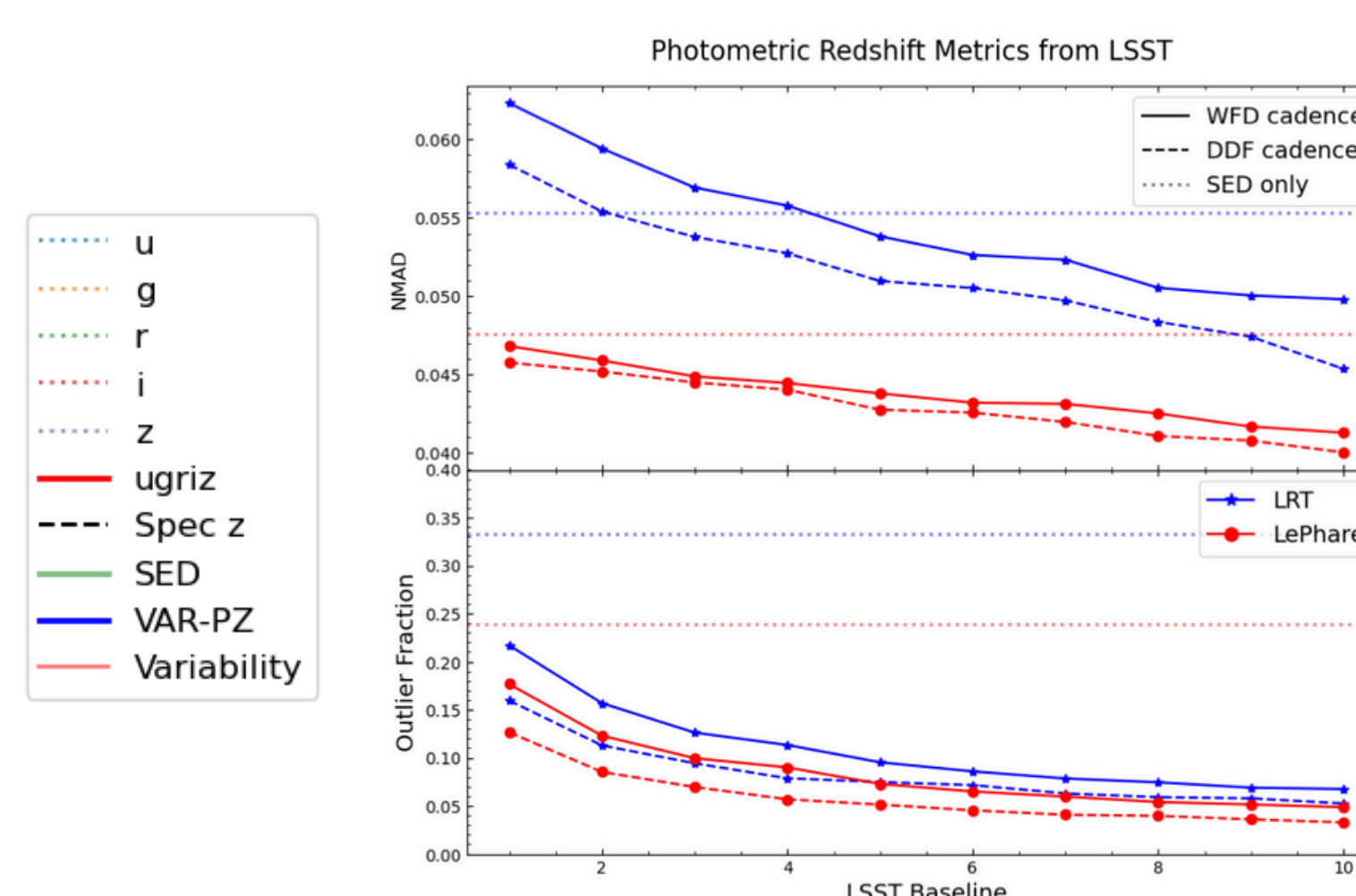


Fig. 5. The evolution of photo-z precision and outlier fraction for our parent sample as a function of the LSST temporal baseline.

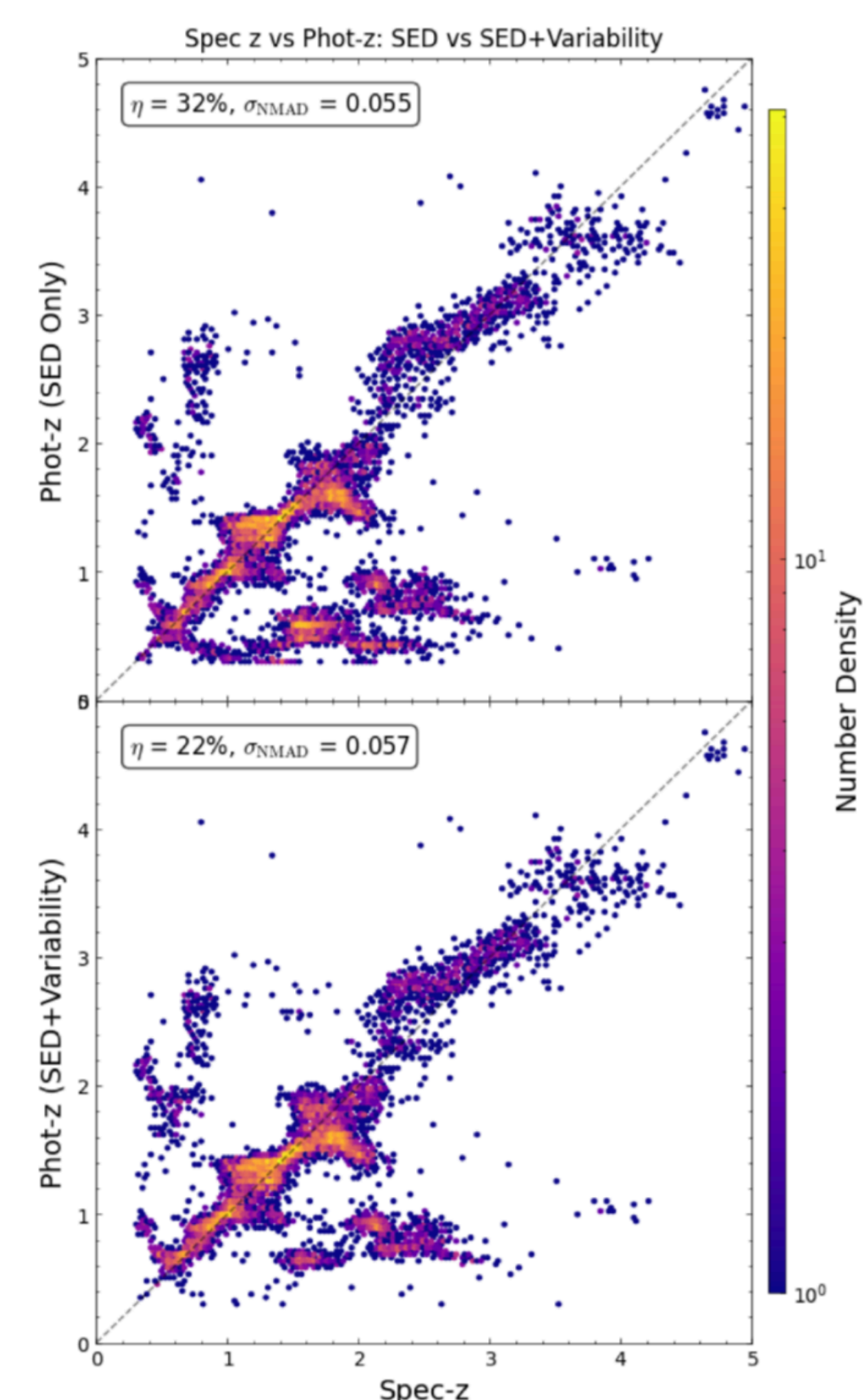


Fig. 6. comparing photometric redshifts derived from SED fitting using LRT, independently (top) and combined with our variability model (bottom), against spectroscopic redshifts.